Hao Li, Ph.D.

Applied Scientist, AWS
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Current Position

Amazon Web Services — Seattle, WA
Applied Scientist II, AWS Rekognition — Oct 2018 - present

- Research and develop machine learning and computer vision algorithms for AWS image and video analytical services, contributed to the launch of new vertical services including *Custom Labels*, *Content Moderation* and *Lookout*. Research on efficient and automatic ML for transfer learning, hyperparameter optimization and neural architecture search.

Research Interests

Machine Learning, Deep Learning, Computer Vision, Distributed Computing

Education

University of Maryland, College Park — College Park, MD
Ph.D. in Computer Science — 2012 – 2018

- Advisors: Prof. Hanan Samet and Prof. Tom Goldstein
- Dissertation: Towards Fast and Efficient Representation Learning
- Committee: Hanan Samet, Tom Goldstein, Larry Davis, David Jacobs, Rama Chellappa

Institute of Computing Technology, Chinese Academy of Sciences — Beijing, China
M.E. in Computer Science and Technology — 2009 – 2012

- Advisor: Prof. Shiguang Shan

Shandong University — Jinan, China
B.E. in Software Engineering — 2005 – 2009

Professional Experience

Department of Computer Science, University of Maryland — College Park, MD
Research / Teaching Assistant — Aug 2012 - Aug 2018

- Visualizing the loss surface of neural networks [NeurIPS'18].
- Understanding training dynamics of quantized neural networks [NeurIPS'17].
- Accelerating CNNs for fast inference [ICLR'17].
- Distributed machine learning with HPC [EuroSys'15, ICML'17].

Machine Learning Team, Nervana Systems (now Intel AI) — San Diego, CA
Algorithms Engineer Intern with Scott Cyphers and Arjun Bansal — May - Aug 2016

- Design and implement the importer module for Neon's graph-backend (Nervana Graph), enabling loading and executing the computation graph of TensorFlow with nGraph.

Machine Learning Department, NEC Labs America — Princeton, NJ
Research Intern / Visiting Student — May - Aug 2015/Jan - May 2016

- Research on the architecture design of Convolutional Neural Networks (CNNs) and filter pruning for the acceleration and compression of CNNs [ICLR'17].

Storage System Department, NEC Labs America Princeton, NJ
Research Intern with Asim Kadav and Erik Kruus May – Aug 2014

– Built MALT, a distributed data-parallel machine learning system with flexible peer-to-peer communication, utilizing one-side RDMA write over InfiniBand [EuroSys'15].

Institute of Computing Technology, Chinese Academy of Sciences Beijing, China
Research Assistant with Prof. Shiguan Shan Jan 2011 - July 2012

– Person tracking with re-identification in non-overlapping camera networks [ACCV'14].

Internet Graphics Group, Microsoft Research Asia Beijing, China
Research Intern with Ying-Qing Xu May – Nov 2010

– Research and develop aesthetic based image search re-ranking.

Media Computing Group, Microsoft Research Asia Beijing, China
Research Intern with Meng Wang Mar – July 2009

– Built a million-scale image/video dataset for web image/video search reranking [ICDMW'09], research on graph-based reranking for web image search [T-IP'12, SIGIR'11].

Publications (Google Scholar, total citations: **2,816**)

1. *Rethinking the Hyperparameters for Fine-tuning*
Hao Li, Pratik Chaudhari, Hao Yang, Michael Lam, Avinash Ravichandran, Rahul Bhotika, Stefano Soatto, International Conference on Learning Representations (**ICLR**), 2020 (citations: 15)

2. *Visualizing the Loss Landscape of Neural Nets*
Hao Li, Zheng Xu, Gavin Taylor, Tom Goldstein
Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2018 (citations: **500+**)

3. *Training Quantized Nets: A Deeper Understanding*
Hao Li*, Soham De*, Zheng Xu, Christoph Studer, Hanan Samet, Tom Goldstein
Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2017 (citations: **100+**)
ICML Workshop on Principled Approaches to Deep Learning, 2017 (**Google Best Student Paper**)

4. *Pruning Filters for Efficient ConvNets*
Hao Li, Asim Kadav, Igor Durdanovic, Hanan Samet, Hans Peter Graf
International Conference on Learning Representations (**ICLR**), 2017 (citations: **1,600+**)
NIPS Workshop on Efficient Methods for Deep Neural Networks (**EMDNN**), 2016

5. *Adaptive Consensus ADMM for Distributed Optimization*
Zheng Xu, Gavin Taylor, **Hao Li**, Mario Figueiredo, Xiaoming Yuan, and Tom Goldstein
International Conference on Machine Learning (**ICML**), 2017 (citations: **35**)

6. *MALT: Distributed Data-Parallelism for Existing ML Applications*
Hao Li, Asim Kadav, Erik Kruus, Cristian Ungureanu
ACM European Conference on Computer Systems (**EuroSys**), 2015 (citations: **50**)
NIPS Workshop on Machine Learning Systems (**LearningSys**), 2015

7. *Streaming News Image Summarization*
Hao Li, Shangfu Peng, Hanan Samet
International Conference on Pattern Recognition (**ICPR**), 2016 (Oral)

8. *Multimodal Graph-Based Reranking for Web Image Search*
Meng Wang, **Hao Li**, Dacheng Tao, Ke Lu, Xindong Wu
IEEE Transactions on Image Processing (**T-IP**), 2012 (citations: **300+**)

9. *Optimizing Multimodal Reranking for Web Image Search*
   Hao Li, Meng Wang, Zhisheng Li, Zheng-Jun Zha, Jialie Shen
   ACM SIGIR Conf. on Research and Development in Information Retrieval (**SIGIR**), 2011 (Short)
10. *MSRA-MM 2.0: A Large-Scale Web Multimedia Dataset*
    Hao Li, Meng Wang, Xian-Sheng Hua
    IEEE ICDM Workshop on Internet Multimedia Mining, 2009 (citations: **100+**)
11. *Simplification and Refinement for Speedy Spatio-temporal Hot Spot Detection Using Spark (GIS Cup).*
    Shangfu Peng, Hong Wei, **Hao Li**, and Hanan Samet
    ACM Int. Conf. on Advances in Geographic Information Systems (**SIGSPATIAL**), 2016 (3rd place)
12. *View Adaptive Metric Learning for Multi-view Person Re-identification*
    Canxiang Yang, Shiguang Shan, Dan Wang, **Hao Li**, Xilin Chen
    Asian Conference on Computer Vision (**ACCV**), 2014

Patents

Passive Pruning of Filters in a Convolutional Neural Network. US Patent. 10,832,136
Parallelized Machine Learning with Distributed Lockless Training. US Patent 9,984,337
Security System Using a Conv. Neural Network with Pruned Filters. US Patent 10,885,437

Awards

Google Best Student Paper Award, ICML 2017 PADL Workshop	2017
Dean's Fellowship, Department of Computer Science, University of Maryland	2012
Stars of Tomorrow Excellent Intern Award, Microsoft Research Asia	2010
Outstanding College Graduate, Shandong Province and Shandong University	2009
Meritorious Winner in Mathematical Contest in Modeling, USA	2008
National Second Prize in China Undergraduate Mathematical Contest in Modeling	2007

Invited Talks

Rethinking the Hyperparameters for Fine-tuning

- Amazon Computer Vision Conference (ACVC), June 2020

Towards Fast and Efficient Representation Learning

- Facebook Research, Menlo Park, CA, May 2018
- IBM Research, Cambridge, MA, May 2018
- Amazon Web Services, Seattle, WA, April 2018
- Apple AI Research, Cupertino, CA, April 2018
- Vicarious, Union City, CA, April, 2018
- Comcast Labs, Washington D.C., Mar. 2018
- Microsoft Research Asia, Bejing, Jan. 2018
- NVIDIA Research, Santa Clara, CA, Nov. 2017

Towards a Deeper Understanding of Training Quantized Neural Networks

- ICML Workshop on Principled Approaches on Deep Learning, Sydney, Aug. 2017

Streaming News Image Summarization.

- ICPR, Cancun, Mexico, Dec. 2016

MALT: Distributed Data-Parallelism for Existing ML Applications.

- EuroSys, Bordeaux, France, April 2015

Teaching Experience

TA, CMSC 764 Advanced Numerical Optimization, with Prof. Tom Goldstein	Spring 2017
TA, CMSC 828Z Reinforcement Learning, with Prof. Hal Daumé III	Fall 2016
TA, CMSC 250H Discrete Structures (Honors), with Prof. Tom Goldstein	Fall 2017
TA, CMSC 216 Introduction to Computer Systems, with Dr. Larry Herman	Fall 2015
TA, CMSC 426 Programming Handheld Systems, with Prof. Atif Memon	Spring 2013
TA, CMSC 330 The Organization of Programming Languages, with Dr. Larry Herman	Fall 2012

Academic Services

Editorial board reviewer for Journal of Machine Learning Research (JMLR)
Reviewer for International Journal of Computer Vision (IJCV)
Reviewer for IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)
Reviewer for IEEE Transactions on Neural Network and Learning Systems (T-NNLS)
Reviewer for IEEE Transactions on Multimedia (T-MM)
Reviewer for IEEE Transactions on Industrial Informatics
Reviewer for Neurocomputing
Reviewer for Multimedia Systems
Reviewer for Neural Information Processing Systems (NeurIPS)
Reviewer for International Conference on Machine Learning (ICML)
Reviewer for International Conference on Learning Representation (ICLR)
Reviewer for IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
Reviewer for International Conference on Computer Vision (ICCV)
Reviewer for European Conference on Computer Vision (ECCV)
Reviewer for ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
Reviewer for AAAI Conference on Artificial Intelligence (AAAI)